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$$\int \frac{x}{\sin^2 x} dx$$

$$\begin{cases} u = x \\ dv = \dfrac{1}{\sin^2 x}\, dx \end{cases} \Rightarrow \begin{cases} du = dx \\ v = -\cot x \end{cases}$$

$$\int \frac{x}{\sin^2 x} dx = -x \cot x - \int (-\cot x) dx = -x \cot x + \int \frac{\cos x}{\sin x} dx$$

$$= -x \cot x + \ln|\sin x| + c$$

$$\text{Want} \int \cot x dx = \int \frac{\cos x}{\sin x} dx$$

$$u = \sin x \Rightarrow du = \cos x dx$$

$$\int \frac{1}{u} du = \ln|u| + c$$

$$\int \cot x dx = \ln|\sin x| + c$$

References